



Aadhaar OKYC API

Technical Specification & Integration Guide

Version 1.0

Document Control

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Version history

Version	Date	Changes / Remarks
1.0	23-Apr-2026	First version of the API. The document has been thoroughly cleaned up and revamped for consistency, accuracy, and readability. It now reflects standardized structure, corrected information, and best practices.

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1. API Overview

1.1. Purpose

TransBnk’s **Aadhaar OKYC API** enables **OTP-based electronic Know Your Customer (e-KYC)** by securely verifying an individual’s Aadhaar details and retrieving **UIDAI-verified demographic and address information** in a structured JSON format.

The API eliminates manual Aadhaar document handling by fetching **authenticated identity attributes**, including name, date of birth, gender, address (split and full), and photograph. Sensitive personal identifiers such as mobile number and email are returned only as **hashed values**, ensuring privacy, UIDAI compliance, and data-minimization.

This service helps lending institutions, fintech platforms, and regulated entities **accelerate customer onboarding**, ensure **regulatory KYC compliance**, reduce fraud risk, and improve turnaround times across digital acquisition journeys.

1.2. Use Cases and Benefits

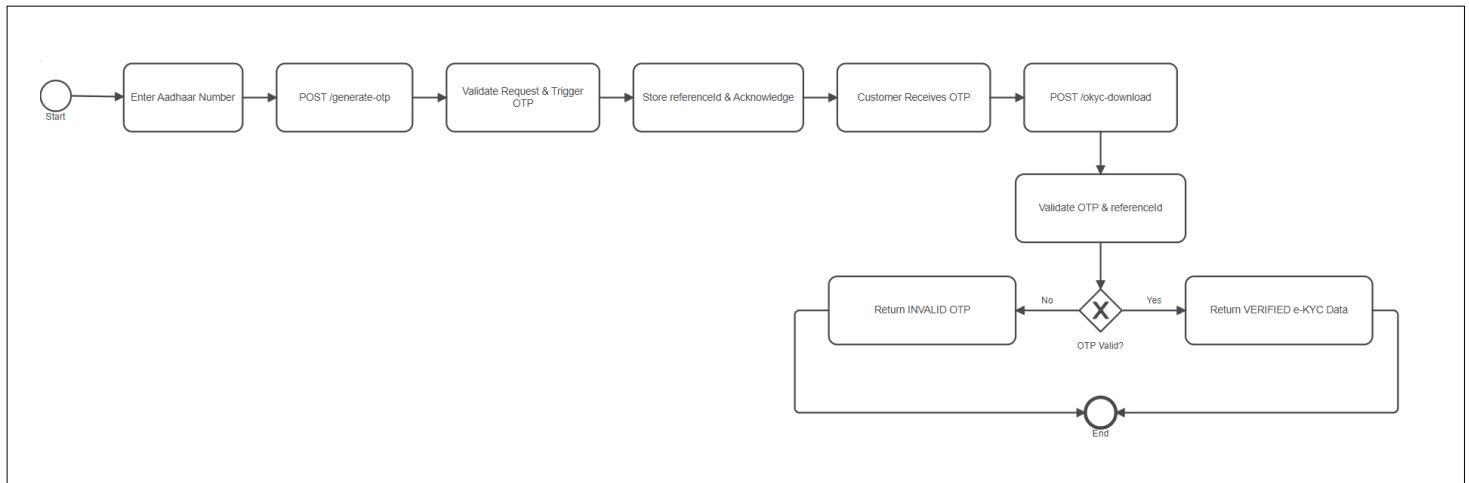
#	Use Cases	Benefits
1	Digital Customer Onboarding	Instantly verifies customer identity using Aadhaar OTP, reducing onboarding time from days to minutes.
2	KYC Compliance for Banks & NBFCs	Provides UIDAI-authenticated demographic data, supporting RBI-compliant KYC and audit-ready processes.
3	Fraud & Identity Risk Prevention	Validates Aadhaar existence and consistency of personal details, reducing impersonation and fake identity risks.

2. Sample Flow Diagram

This flow illustrates the **end-to-end lifecycle** of how a client system interacts with the **TransBnk Aadhaar OKYC API ecosystem**—from initiating Aadhaar OTP generation to downloading verified e-KYC details.

The process demonstrates how TransBnk performs Aadhaar verification through **two core APIs**:

1. **Aadhaar OTP Generation API**
2. **Aadhaar OKYC Download API**



3. API usage instructions

3.1. Base URLs

#	API Name	UAT URL	Production URL
1	Aadhaar OTP Generation API	https://sandbox-api.trusthub.in/generate-otp	https://api.trusthub.in/generate-otp
2	Aadhaar OKYC Download API	https://sandbox-api.trusthub.in/okyc-download	https://api.trusthub.in/okyc-download

3.2. Prerequisites

3.2.1. Security

- Provide the list of IP addresses for whitelisting.
- Ensure all HTTPS requests use TLS 1.2.
- Complete Sign-up, Onboarding, and Activation on TransBnk's APICMS Portal.
- Avoid sending special characters in API requests.
 - Special characters are blocked at the API Gateway for security hardening.
 - If encountered, replace them with a space.

3.2.2. Required Headers

#	Key	Value / Description
1	x-api-key	The API key will be provided by TransBnk after activation and IP whitelisting.
2	Content-Type	application/json

3.3. HTTP Method

#	API Name	HTTP Method
1	Aadhaar OTP Generation API	POST
2	Aadhaar OKYC Download API	POST

3.4. API Specifications

3.4.1. Aadhaar OTP Generation API

The Aadhaar OTP Generation API initiates the Aadhaar Online KYC (OKYC) process by securely submitting the customer's Aadhaar number to trigger an OTP delivery to the Aadhaar-registered mobile number.

This API generates a unique `referenceId`, which acts as a transaction handle for subsequent OTP validation and e-KYC data retrieval. It ensures explicit customer consent and enables regulated entities to perform UIDAI-aligned identity verification in a fully digital and paperless manner.

Request Body Format & Example

Category	Request Format	Request Example
Data Exchange Format	JSON	JSON
Working request	<pre>{ "input": { "aadhaarNumber": "string" } }</pre>	<pre>{ "input": { "aadhaarNumber": "47796228XXXX" } }</pre>

Request Parameters

Field Name	Mandatory	Data Type	Description
input	Yes	Object	Wrapper object that encapsulates all request parameters required for initiating the Aadhaar OTP generation process.
aadhaarNumber	Yes	String	The 12-digit Aadhaar number of the customer for whom OTP-based Aadhaar OKYC is to be initiated. This value must be provided as a numeric string and is used to trigger a One-Time Password (OTP), which is sent to the mobile number registered with Aadhaar. The OTP serves as explicit customer consent for identity verification and is valid for 10 minutes from the time of generation. Values that are blank, null, non-numeric, not exactly 12 digits in length will result in a validation failure.400 (Bad Request)

Response Body Format & Example

Category	Response Format	Response Example
Data Exchange Format	JSON	JSON
Working Response :	<pre>{ "status": "string", "requestId": "string", "type": "string", "serviceStatusCode": "number", "data": { "referenceId": "string" }, "vendorResponse": [{ "statusCode": "number", "name": "string", "sequence": "number" }] }</pre>	<pre>{ "status": "success", "requestId": "98cfca06-fd94-49e5-aab9-162d673e4010", "type": "point", "serviceStatusCode": 200, "data": { "referenceId": "75238564" }, "vendorResponse": [{ "statusCode": 200, "name": "VermilionSwift", "sequence": 1 }] }</pre>

Response Parameters

Field Name	Data Type	Description
status	String	Overall outcome of the API request. Indicates whether the request was successfully processed (e.g., success).
requestId	String	Unique identifier for tracking the API request across systems and logs.
type	String	Indicates the billing or service invocation type associated with the request (e.g., point).
serviceStatusCode	Number	HTTP-equivalent status code representing the processing result of the API at the service level.
data	Object	Contains the functional response payload for the API call. For Aadhaar OTP generation, this object includes the reference identifier required for downstream OKYC operations.
data.referenceId	String	Unique reference ID generated for the Aadhaar OTP transaction. This ID must be passed to the Aadhaar OKYC Download API along with the OTP.

3.4.2. Aadhaar OKYC Download API

The Aadhaar OKYC Download API is used to validate the OTP and retrieve verified Aadhaar e-KYC details for a previously initiated Aadhaar OTP request.

Businesses and financial institutions invoke this API to:

- Confirm whether the Aadhaar OTP validation was successful or failed
- Retrieve UIDAI-verified demographic and address attributes
- Complete digital KYC onboarding using a consent-driven flow

This API must be called using the referenceId generated by the Aadhaar OTP Generation API along with the customer-provided OTP.

Request Body Format & Example

Category	Request Format	Request Example
Data Exchange Format	JSON	JSON
Working request: Success	<pre>{ "input": { "referenceId": "string", "otp": "string" } }</pre>	<pre>{ "input": { "referenceId": "75238564", "otp": "883880" } }</pre>

Request Parameters

Field Name	Mandatory	Data Type	Description
input	Yes	Object	Wrapper object containing parameters required to validate the Aadhaar OTP and download Aadhaar e-KYC details.
referenceId	Yes	String	Unique 8-digit numeric reference ID generated during the Aadhaar OTP Generation API call. This identifier is used to correlate and link the OTP verification request with the original Aadhaar OTP transaction and must be passed unchanged. Values that are blank, null, non-numeric, not exactly 8 digits in length will result in a validation failure.
otp	Yes	String	One-Time Password (OTP) received on the mobile number registered with Aadhaar. This value is used to authenticate explicit customer consent and complete the Aadhaar OKYC verification process. The OTP must be provided as a numeric string consisting of exactly 6 digits. Values that are blank, null, non-numeric, not exactly 6 digits in length will result in a validation failure.

Response Body Format & Example

Category	Response Format	Response Example
Data Exchange Format	JSON	JSON
Working Response : Success	<pre>{ "status": "string", "requestId": "string", "type": "string", "serviceStatusCode": "number", "data": { "status": "string", "message": "string", "careOf": "string", "fullAddress": "string", "dob": "string", "emailHash": "string", "gender": "string", "name": "string", "splitAddress": { "country": "string", "district": "string", "house": "string", "landmark": "string", "pincode": "string", "postOffice": "string", "state": "string", "street": "string", "subdistrict": "string", "vtc": "string" }, "yearOfBirth": "string", "mobileHash": "string", "photo": "string", "shareCode": "string" }, "vendorResponse": [{ "statusCode": "number", "name": "string", "sequence": "number" }] }</pre>	<pre>{ "status": "success", "requestId": "89437894-f1bc-451f-b276-17ec369f0468", "type": "point", "serviceStatusCode": 200, "data": { "status": "VALID", "message": "Aadhaar Card Exists", "careOf": "", "fullAddress": "VILLA, NEAR ANIKET HOSPITAL, Mumbai, Maharashtra, India, 400042", "dob": "25-09-1988", "emailHash": "1260a577cb27e5bc47061ea03b51d0a5b9e8a99458afe539e7a18512575319b1", "gender": "M", "name": "John Ken Doe", "splitAddress": { "country": "India", "district": "Mumbai", "house": "VILLA", "landmark": "NEAR ANIKET HOSPITAL", "pincode": "400042", "postOffice": "Bhandup East", "state": "Maharashtra", "street": "", "subdistrict": "Mumbai", "vtc": "Mumbai" }, "yearOfBirth": "1988", "mobileHash": "8e370ebecdc6ea316a999e60dc0a43cf39ea56acd66e6f8e5ed3e4cb885a1168", "photo": "/9j/4TEf/Z", "shareCode": "2345" }, "vendorResponse": [{ "statusCode": 200, "name": "VermilionSwift", "sequence": 1 }] }</pre>

Response Parameters

Field Name	Data Type	Description
status	String	Indicates overall request status (success or error).
requestId	String	Unique identifier generated by TransBnk to track the API request for logging, monitoring, and audit purposes.
type	String	Indicates the service invocation or billing type associated with the request (for example, point).
serviceStatusCode	Number	HTTP-equivalent status code representing the final processing status of the request at the service layer (e.g., 200 for success).
data	Object	Contains the Aadhaar OKYC response payload with verified demographic, address, and identity attributes. Note : All fields in this object of type string will have a default value of an empty string "" if data is not available.
data.status	String	Aadhaar OKYC validation result. Indicates whether the OTP verification was successful. Validation result of the Aadhaar (VALID indicates the Aadhaar exists and is verified).
data.message	String	Human-readable message describing the result (e.g., "Aadhaar Card Exists").
data.careOf	String	Guardian or relationship field (e.g., S/O, D/O, W/O).
data.fullAddress	String	Complete address in a single string format.
data.dob	String	Date of birth of the Aadhaar holder in DD-MM-YYYY format.
data.emailHash	String	SHA-256–based hashed value of the user’s email (used for verification, not plain text). Refer section “Hash Logic for Mobile & Email Verification”
data.gender	String	Gender of the individual (M, F, or others).
data.name	String	Full name as per Aadhaar records.
data.splitAddress	Object	Structured representation of the Aadhaar address, broken down into individual address components.
data.splitAddress.country	String	Country associated with the Aadhaar address.
data.splitAddress.state	String	State associated with the Aadhaar address.
data.splitAddress.district	String	District associated with the Aadhaar address.
data.splitAddress.subdistrict	String	Sub-district or taluka associated with the Aadhaar address.
data.splitAddress.vtc	String	Village, town, or city name associated with the Aadhaar address.
data.splitAddress.house	String	House number or building details from the Aadhaar address.

Field Name	Data Type	Description
data.splitAddress.street	String	Street or road information from the Aadhaar address, if available.
data.splitAddress.landmark	String	Landmark details provided in the Aadhaar address.
data.splitAddress.postOffice	String	Post office corresponding to the Aadhaar address.
data.splitAddress.pincode	String	Postal PIN code associated with the Aadhaar address.
data.mobileHash	String	SHA-256–based hashed value of the registered mobile number. Refer section “ Hash Logic for Mobile & Email Verification”
data.photo	String	Base64-encoded image (may be empty if not provided).
data.shareCode	String	A user-defined or system-provided code used for decrypting and verifying sensitive data.

Hash Logic for Mobile & Email Verification

The API does not return the actual mobile number or email. Instead, it provides hashed values (**mobileHash** and **emailHash**) for privacy and security. To verify user-provided input, follow the process below:

Step 1: Collect Input

Ask the user for their mobile number or email (plain text).

Step 2: Apply Hash Formula

SHA-256(SHA-256(Input + ShareCode)) repeated n times

Step 3: Determine Iteration Count (n)

Take the last digit of the Aadhaar number:

- If the last digit is 0 or 1 → perform hashing once
- If the last digit is 2–9 → perform hashing n times
(e.g., last digit = 4 → hash 4 times)

Step 4: Compare Hash

Compare the final computed hash with:

- mobileHash (for mobile verification), or
- emailHash (for email verification)

If both hashes match, the input is verified successfully.

4. Error Handling

4.1. API Gateway Error Formats

#	Error Scenario	Description	Sample Error Response
1	Missing/Incorrect API Key or wrong method (non-POST)	Requests missing an API key or containing an incorrect API key are rejected. Using an incorrect HTTP method results in a	{ "message": "Forbidden" }

		'Forbidden' response.	
2	Empty or incorrect Request Body	The payload is missing or malformed.	{ "message": "Invalid request body" }
3	IP not whitelisted	The caller's IP address is not authorized to access the API.	{"Message": "User: anonymous is not authorized to perform: execute-api:Invoke on resource..."}

4.2. Errors

Note :- In response "code" field appears for errors below are the details

#	Error Scenario	Description	Sample Error Response
1	Aadhaar Number Not Registered (Valid format but does not exist)	When a valid Aadhaar number format is submitted but the number itself is not registered in the system, the API responds with code 400 and message "Invalid Input". The details field remains empty, and the overall status is shown as 200, indicating that the request was technically processed but rejected due to the non-existent Aadhaar number.	{ "code": 400, "message": { "message": "Invalid Input", "details": {} }, "status": 200 }
2	OTP Not Found (Valid format but non-existent)	If an OTP is provided in the correct format but does not exist in the system (for example, it has expired), the API returns code 400 with message "Invalid Input". The details field is empty, and the overall status is 200, meaning the request was received but the OTP could not be validated.	{ "code": 400, "message": { "message": "Invalid Input", "details": {} }, "status": 200 }
3	Reference ID Not Found (Valid format but non-existent)	When a valid reference ID format is submitted but no matching record exists, the API returns a wrapper response with status "success" and serviceStatusCode 200. However, the data payload contains the message "Invalid reference_id" and all other fields are empty, confirming that the reference ID is invalid. The vendor response also shows statusCode 200, but it reinforces that the reference ID does not exist in the system.	{ "status": "success", "requestId": "9f0c0a2b-8d3f-46e9-bc91-5014f8684d6f", "type": "point", "serviceStatusCode": 200, "data": { "status": "", "message": "Invalid reference_id", "careOf": "", "fullAddress": "", "dob": "", "emailHash": "", "gender": "", "name": "", "splitAddress": { "country": "", "district": "", "house": "", "landmark": "", "pincode": "", "postOffice": "", "state": "", "street": "", "subdistrict": "", "vtc": "" }, "yearOfBirth": "", } }

			<pre> "mobileHash": "", "photo": "", "shareCode": "" }, "vendorResponse": [{ "statusCode": 200, "name": "VermilionSwift", "sequence": 1 }] } </pre>
4	Limit Exceeded (multiple OTP generation attempts for identical inputs)	If multiple OTP generation requests are made with the same inputs beyond the allowed threshold, the API enforces rate limiting. In this case, the response includes code 429 with message "Too Many Requests". The details field is empty, and the overall status is 200, indicating that the request was processed but blocked due to exceeding the permitted number of OTP attempts.	<pre> { "code": 429, "message": { "message": "Too Many Requests", "details": {} }, "status": 200 } </pre>

5. Support

Please refer to the escalation matrix below for raising any queries related to the API documentation content in this document.

Level	Environment	Support Email ID
1st level	UAT/Production	support@transbnk.co.in